

ASIRI SANDAKELUM

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🌐 linkedin.com/in/asiri-sandakelum | 📁 Asiri Sandakelum Portfolio

Professional Summary

I am a Mechanical Engineer specialising in mechanical design, FEA, and manufacturing systems. I bridge the gap between advanced 3D simulation and physical production, ensuring designs are robust and manufacturable. With expertise in industrial project management and facility optimization, I am actively seeking full-time roles to drive scalable engineering solutions.

Education

University of Moratuwa

BSc Eng (Hons) specialised in the field of Mechanical Engineering.

Moratuwa, Sri Lanka

2022 – Present

- **Relevant Coursework:** Manufacturing Processes, Machine Design, Mechatronics & PLC, Embedded Systems, Engineering Mathematics, Thermodynamics, Fluid Mechanics, Engineering Materials, Operations Research

Work Experience

Future Fibres Lanka (Pvt) Ltd — Maintenance Section

Mechanical Engineering Intern

Biyagama EPZ, Sri Lanka

Dec 2024 – May 2025

- Redesigned a **Bobbin Winding Tensioner** in SolidWorks, eliminating roller misalignment; improved yarn tension consistency and reduced thread breakage, verified through fabrication and live testing.
- Developed an **IoT-based Temperature & Humidity Monitoring System** (ESP32, DHT22, DS18B20) with real-time cloud logging via ThingSpeak, enabling remote environmental tracking across production zones.
- Conducted a carbon dust filtration case study for the AeroSIX plate cutting area; engineered a sealed HEPA filtration enclosure that significantly improved air quality and compliance with worker safety standards.
- Drafted AutoCAD layout drawings for floor plans, piping systems, and drainage; supported CMMS implementation and contributed to preventive maintenance scheduling and 5S standardisation programmes.
- Gained direct exposure to carbon fibre composite manufacturing processes for high-performance rigging systems distributed to global marine and sailing markets.

Engineering Projects [🔗 View Full Portfolio](#)

Active Knee Exoskeleton for Osteoarthritis Treatment

Ongoing

- Designed a wearable active exoskeleton targeting Knee Adduction Moment (KAM) reduction for mild-to-moderate OA patients; optimised a polycentric 5-Bar Linkage via Genetic Algorithm (GA), achieving an ICR tracking error of **0.47 mm**.
- Implemented remote cable-driven actuation with FSM-based gait phase detection and impedance control; target assistive torque: **25 Nm**.
- **Tools:** MATLAB, Genetic Algorithm, SolidWorks

Below-Knee Prosthetic Leg — Fatigue Load Testing Mechanism

Dec 2025

- Designed a fully mechanical gait-simulation mechanism (heel-strike → stance → toe-off → swing) for prosthetic fatigue testing under vertical loads up to **100 kg**; selected optimal crank-link mechanism via weighted decision matrices.
- **Tools:** SolidWorks, DFMA Principles, 3D Printing (PLA)

Operational Throughput Enhancement — Abans Auto (Pvt) Ltd

Ongoing

- Identified bottlenecks in PDI/QC (75 min/unit) and batch material handling (90 min/batch) at a Hero Motorcycle Assembly Plant; proposed drive-through workstation layout, dedicated functional test track, and strategic safety stock management.

Reverse Engineering — Bicycle Disc Brake System

May 2024

- Performed full material characterisation (burn tests, density measurements, Rockwell hardness testing); developed optimised 3D models and validated performance via FEA simulation and generative design.
- **Tools:** Fusion 360, ANSYS FEA, Generative Design

Gearbox Design — Combat Robot Weapon Systems

Nov 2024

- Designed a high-torque multi-speed gearbox for vertical and horizontal spinner weapons; completed gear selection, shaft stress analysis, bearing selection, and power transmission calculations from first principles.
- **Tools:** Solid Edge, Machine Design Principles

Smart Medibox — IoT Embedded System

Aug 2024

- Built an IoT-connected medicine box monitoring temperature, humidity, and light levels with automated alerts and a real-time Node-RED dashboard; simulated and validated on Wokwi.
- **Tools:** C++, Node-RED, MQTT, Wokwi

- Led mechanical design and systems integration for a fully autonomous car wash robot; interfaced Arduino Mega and ESP8266 with a custom MIT App Inventor mobile application.
- *Tools:* SolidWorks, Arduino, MIT App Inventor

Production Line Optimisation — Aruna Tea Factory

Nov 2024

- Identified throughput bottlenecks in withering and packing sections; proposed equipment modifications and rescheduling strategies validated through simulation-based analysis.

Leadership & Volunteering

Department of Mechanical Engineering, University of Moratuwa

Student Representative

2025 – Present

- Served as the primary liaison between the department administration and student body; negotiated academic schedules and resolved grievances affecting student welfare.
- Spearheaded coordination of departmental career fairs and industry networking events; developed a centralised communication system achieving **100% information reach** across the cohort.
- Led planning and budgeting for extracurricular events and professional development workshops, managing cross-functional student committees.

IEEE Robotics & Automation Society (RAS), University of Moratuwa

Vice Chairman

Nov 2024 – Oct 2025

- Directed executive decision-making and annual strategic planning for one of the university’s most active IEEE student chapters.
- Mentored junior committee members in project management, event planning, and technical execution.

Co-Chair, BatTalk 2.0

Mar 2024 – Oct 2024

- Organised a flagship technical talk series on AI/ML career dynamics, coordinating with industry professionals and university faculty; managed end-to-end logistics and speaker outreach.

Sasnaka Sansada Foundation

District Coordinator — Nuwara Eliya

Mar 2024 – Dec 2024

- Managed district-level outreach operations, volunteer recruitment, and provincial seminar logistics.

Ganitha Sawiya Coordinator

May 2022 – May 2023

- Organised **50+ mathematics seminars** and educational workshops across the Nuwara Eliya District, directly improving O/L and A/L mathematics pass rates among rural school students.

EXMO 2023 — University of Moratuwa

Exhibit Coordinator

Apr 2023 – Jul 2023

- Curated and managed mechanical engineering exhibits for Sri Lanka’s premier university engineering exhibition; oversaw space allocation and technical safety for thousands of visitors.

Technical Skills

CAD & Design: SolidWorks, Fusion 360, Solid Edge, AutoCAD

Simulation & Analysis: ANSYS (FEA), MATLAB, Genetic Algorithm Optimisation, DFMA

Programming: Python, C/C++, Arduino IDE

IoT & Embedded Systems: ESP32, DHT22, DS18B20, ThingSpeak, Node-RED, MQTT

Manufacturing: Basic manufacturing processes, CNC machining, sheet metal fabrication, welding, 3D printing

Engineering Software: MS Office Suite, Photoshop, DaVinci Resolve, After Effects

Languages: English (Professional Proficiency), Sinhala (Native)

Core Competencies: Mechanical design, preventive maintenance, process optimisation, technical documentation, cross-functional team leadership

References

Dr. Pubudu Ranaweera

Senior Lecturer

Department of Mechanical Engineering

University of Moratuwa, Sri Lanka

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